

Lesson 5 · Mixed Practice

Learning intention

Choose the right calculus method from the wording of a question, and write Merit and Excellence answers under time pressure.

Success looks like

- I can name the method before I start writing.
- I can link two steps and justify a maximum in a sentence.
- I can build a formula from words and optimise it.

Do Now

1 Differentiate $y = (3x - 1)(x + 2)$.

2 Find the turning point of $y = 2x^2 - 12x + 7$ and state its nature.

3 $f'(x) = 8x - 3$ and $f(1) = 6$. Find $f(x)$.

4 $f'(x)$ changes from negative to positive at $x = -2$. What happens on the graph of f at $x = -2$?

The one idea

read the words $\longrightarrow$ **name the method** $\longrightarrow$ **write the calculus**

“Use calculus” means the derivative or anti-derivative must be in your working.

A calculator answer alone shows no method.

Which method?

Name the method for each. Do not solve.

1. Find the gradient of the curve at $x = 4$.
2. Find the equation of the tangent at $x = -1$.
3. Find the maximum height of the ball.
4. The curve passes through $(2, 5)$; find $f(x)$ given $f'(x)$.
5. For what values of x is f decreasing?
6. Given the acceleration, find how far the car travels.

Practice

- 1 Find the gradient of $y = 2x^3 - 5x + 4$ at the point where $x = -2$.
- 2 $f'(x) = 12x^2 - 6$ and $f(1) = 3$. Find $f(x)$.
- 3 A particle has velocity $v = 6t - t^2 \text{ m s}^{-1}$. Find its acceleration at $t = 2$.

Merit: link two steps

Two unknowns $\rightarrow$ two equations. A value given $\rightarrow$ solve for the time first.

4 The curve $y = ax^3 + bx$ has a turning point at $(1, -4)$. Find a and b .

equation 1: the point is on the curve

equation 2: the gradient there is 0

Practice

- 1 The curve $y = x^2 + px + q$ has a turning point at $(2, 5)$. Find p and q .
- 2 The number of visitors at a festival is $V = 600t - 25t^2$ after t hours ($0 \leq t \leq 24$). Find the rate of change of visitors when $V = 3200$.
- 3 Find the maximum value of $f(x) = 3 + 12x - 2x^3$ for $x \geq 0$, and justify that it is a maximum.

Excellence: build the model

Write it in this order

1. Name the variables.
 2. Constraint $\rightarrow$ one variable.
 3. Differentiate, $= 0$, solve; reject impossible values.
 4. Justify; answer in a sentence with units.
- 5** An open box has a square base of side x cm and height h cm. Its outside surface area is 108 cm^2 . Find the maximum volume.

Practice

1 A 48 cm piece of wire is cut into two parts. Each part is bent into a square. Where should the wire be cut so that the total area of the two squares is a minimum? Justify that it is a minimum.

2 For $y = x^3 - 3a^2x$, where $a > 0$ is a constant, show that the turning points are at $x = \pm a$, and find the difference between the maximum and minimum y -values in terms of a .

Ways to lose marks

- L1 Write the derivative before substituting.
- L2 Rates from the derivative; values from the original.
- L3 A point needs y ; answer the quantity asked.
- L4 Justify every maximum and minimum.
- L6 Meaning and units for every rate.

Summary

1. Name the method from the wording first.
2. Merit: two conditions, two equations; value given, solve for t .
3. Excellence: variables, constraint, one variable, optimise, justify.
4. Every answer: working, sentence, units.

Practice sheet · 30 minutes, timed

Exam conditions: no notes, calculator allowed.

- Question 1: four Achieved parts, about 12 minutes.
- Question 2: two Merit parts, about 10 minutes.
- Question 3: Excellence, about 8 minutes.

Attempt every part; write the derivative first.

Homework

Homework sheet: 9 questions, about 60 minutes

- Questions 1 to 6: mixed Achieved and Merit revision.
- Questions 7 and 8: Excellence exam questions.
- Question 9: Extension, a general proof.

Optional extra: redo any earlier practice sheet question you got wrong.